HomePulse



Term Project Proposal

Household electricity prediction and benchmarking

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Data source: EIA RECS

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Household electricity prediction and benchmarking

PROBLEM

Users need a simple way to estimate electricity use and understand differences by home type, size and location.

AUDIENCE

Homeowners seeking an initial estimate and residential energy analysts comparing homes.



U.S. Energy Information Administration

RECS 2020 / U.S. households

PROPOSED SOLUTION

A clearer picture of household electricity

01 Explore

Explore electricity use by home type, size and location.

02 Estimate

Use home characteristics and location to estimate annual kWh.

03 Understand

Compare home types and regions, and see influential inputs.



APPROACH AND ARCHITECTURE

System architecture

MODEL DEVELOPMENT

U.S. Energy Information Administration

RECS 2020 CSV

Household characteristics + KWH

Clean with pandas

SQLite database

Clean household data for training and household comparisons

Training pipeline

Baseline + XGBoost
Tune and evaluate before saving

USER APPLICATION

Streamlit interface

Enter home details and location.
View estimates and comparisons.

Prediction service

Validate inputs, apply preprocessing and return annual electricity kWh.

Saved model pipeline

Trained model and fitted preprocessing stored together.

PROPOSED TECHNOLOGIES

pandas
Data cleaning

scikit-learn
Prepare and evaluate

XGBoost
Electricity prediction

Plotly
Interactive charts

Streamlit
App interface

Scope, risks and responses

CORE SCOPE

Use home characteristics and location to estimate annual electricity consumption and compare home types and regions.

WHAT COULD GO WRONG?	HOW I WILL ADDRESS IT
Data quality	Check missing values, unusual records and survey codes.
User adoption	Keep inputs simple and results easy to understand.

PLAN AND SUCCESS CRITERIA

Way forward

STEP 1	**Data and baseline**	Audit variables and missing codes. Explore patterns and establish baseline.
STEP 2	**Prediction pipeline**	Train and tune XGBoost. Evaluate on untouched test households.
STEP 3	**App and demonstration**	Build dashboard and explanations. Check inputs and document reproduction.

SUCCESS **Evaluate against the baseline, report test errors and limitations, and demonstrate a reproducible end-to-end prediction.**

Data: EIA Residential Energy Consumption Survey (RECS), 2020